

# End-to-End Testing Guide

Step-by-step test plans for every module — from patient signup to full clinical flow

## What this document covers

- Section 1** Environment Setup & Test Credentials
- Section 2** Authentication — signup, login, role-based redirect
- Section 3** Patient Journey — register !' book !' consult !' receive prescription
- Section 4** Doctor / HMS Flow — OPD !' clinical notes !' prescription !' discharge
- Section 5** Spine AYUSH — assess !' protocol !' treat !' track outcomes
- Section 6** Shop & Checkout — cart !' Razorpay !' order confirmation
- Section 7** Admin & Super Admin — roles, data, platform management
- Section 8** Notifications — DB writes + doctor alert verification
- Section 9** Security, Performance & Accessibility
- Section 10** Database Verification Checklist — what to check after each test
- Section 11** Running Automated Tests (Playwright)
- Section 12** Bug Report Template

### Testing philosophy

Test the FLOW, not just the button. Every test in this guide ends with a database verification step — confirming the data was actually saved to the correct tables and notifications were triggered. A green UI is not sufficient; the backend must be verified.

## Section 1 — Environment Setup & Test Credentials

Before running any test, set up two environments: Manual (browser) and Automated (Playwright). Both need the test credentials below.

### 1.1 Test User Accounts (already exist in your DB)

| Role                   | Email                      | Access level                   | Use for testing               |
|------------------------|----------------------------|--------------------------------|-------------------------------|
| Super Admin (Owner)    | curesure4u@gmail.com       | Everything incl. owner pages   | Owner-tier tests, all modules |
| Admin (Dr Sindhu Roja) | alshifachennaidr@gmail.com | All modules except owner pages | Admin flow, data building     |
| Doctor                 | jawahirasaleem@gmail.com   | HMS doctor modules             | Doctor prescription, OPD flow |
| Patient                | curesure4u@gmail.com       | Patient dashboard, booking     | Booking, shop, profile        |

#### Password note

Use the real passwords for these accounts. Never put real passwords in code or commit to git. Store them only in .env.e2e (gitignored).

### 1.2 Create the .env.e2e file (for Playwright)

Create this file at the root of the project. It is gitignored — never commit it.

```
E2E_BASE_URL=https://ayuzee.com
E2E_PATIENT_EMAIL=curesure4u@gmail.com
E2E_PATIENT_PASSWORD=<your_password>
E2E_DOCTOR_EMAIL=jawahirasaleem@gmail.com
E2E_DOCTOR_PASSWORD=<your_password>
E2E_SUPABASE_URL=https://gsjkrmrmlxakcecpfiz.supabase.co
E2E_SUPABASE_ANON_KEY=<anon_key_from_supabase_dashboard>
```

### 1.3 Supabase Dashboard — what to have open

- Table Editor — to verify rows are being written correctly after each test.
- Authentication ! Users — to verify new user registrations.
- SQL Editor — for the database verification queries in Section 10.
- Logs ! Edge Functions — to check webhook/notification edge functions.

### 1.4 Test devices & browsers

| Test type         | Browser/Device                                           | How to run                           |
|-------------------|----------------------------------------------------------|--------------------------------------|
| Manual functional | Chrome (latest)                                          | Open ayuzee.com, follow test steps   |
| Mobile responsive | Chrome DevTools ! iPhone 12 ! Toggle device, set 390×844 |                                      |
| Automated E2E     | Chromium (Playwright)                                    | node scripts and npx playwright test |
| Accessibility     | Chrome + axe DevTools extension                          | Run axe scan on each page            |

## Section 2 — Authentication Testing

Authentication is the gateway to everything. Test all four paths: new signup, existing login, wrong password, and role-based redirect.

### 2.1 New Patient Signup



| Step | Action                                                                        | Expected result                                                  | DB table to verify                 |
|------|-------------------------------------------------------------------------------|------------------------------------------------------------------|------------------------------------|
| 1    | Go to ayuzee.com/auth?mode=signup                                             | Signup form visible with fullName, phone, email, password fields | -                                  |
| 2    | Fill: name=Test Patient, phone=9999999999, email=<new>, password=TestPass!234 | Fields accept input correctly                                    | -                                  |
| 3    | Click Sign Up button                                                          | Toast: 'Check your email' OR redirect to dashboard               | auth.users — 1 new row             |
| 4    | Open email, click confirmation link                                           | Browser opens ayuzee.com, session created                        | auth.users: email_confirmed_at set |
| 5    | Check redirect destination                                                    | Lands on /dashboard or /patient                                  | profiles — new row created         |
| 6    | Verify role assigned                                                          | User has 'patient' role                                          | user_roles — role='patient'        |

### 2.2 Existing User Login — all roles

| Role        | Login email                | Expected redirect      | What to verify                           |
|-------------|----------------------------|------------------------|------------------------------------------|
| Patient     | curesure4u@gmail.com       | /dashboard or /patient | Patient dashboard loads, no HMS visible  |
| Doctor      | jawahirasaleem@gmail.com   | /vaidya or /doctor     | Doctor tools visible, patient list shows |
| Admin       | alshifachennaidr@gmail.com | /hms or /admin         | All HMS modules accessible in sidebar    |
| Super Admin | curesure4u@gmail.com       | /hms with owner pages  | Spine Super Admin page accessible        |

### 2.3 Negative tests

- Wrong password! Toast shows 'Invalid credentials' or similar error. User stay page.
- Non-existent email! Same error toast. No account creation side-effect.
- Empty form submit! HTML5 / toast validation fires. No API call made.
- SQL injection in email field (e.g. ' OR 1=1--)! Treated as invalid email format

#### Role-based access gate test

After logging in as Patient, manually navigate to /hms — you should be redirected away or see 'Access denied'. The HMS modules are invisible in the sidebar. This confirms the role guard is working.

## Section 3 — Patient Journey (End-to-End)

The complete patient flow: discover the clinic !' book appointment !' receive consultation !' health. Each step must write to the correct DB tables.

### 3.1 Find a Doctor & Book Appointment

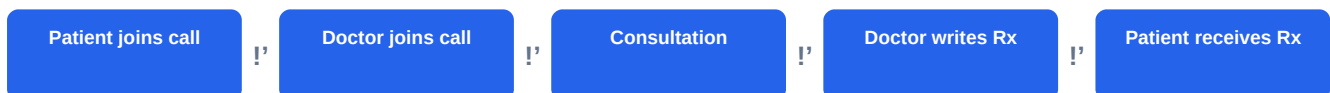


| Step | Action                                        | Expected UI result                            | DB write to verify                            |
|------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| 1    | Login as patient (curesure4u@gmail.com)       | Redirects to patient dashboard                | auth.users: last_sign_in_at updated           |
| 2    | Navigate to /doctors                          | Doctor listing page with filter/search        | doctors table — 3 rows exist                  |
| 3    | Click a doctor card !' View                   | Doctor detail: bio, fee, slots, ratings       | —                                             |
| 4    | Click 'Book Consultation'                     | BookingDialog opens with date/time picker     | —                                             |
| 5    | Select mode: Video / In-clinic                | Mode button highlights                        | —                                             |
| 6    | Select an available time slot                 | Slot highlights, confirm button activates     | —                                             |
| 7    | Add note: 'Lower back pain for 3 months'      | Note text accepted                            | —                                             |
| 8    | Click 'Pay & Book' (if Razorpay) or 'Confirm' | Razorpay checkout opens OR booking confirmed  | appointments — 1 new row                      |
| 9    | After booking, check dashboard                | Appointment shows as 'Pending' or 'Confirmed' | appointments: status='pending'/'confirmed'    |
| 10   | Doctor notified                               | Doctor sees new appointment in queue          | Check follow_up_reminders or doctor dashboard |

### 3.2 Patient Profile & Health Records

| Test                  | Steps                                          | DB table to verify                                             |
|-----------------------|------------------------------------------------|----------------------------------------------------------------|
| Upload profile photo  | Go to /patient/profile !' Upload               | profiles: avatar_url updated; Supabase Storage: avatars bucket |
| Update health info    | Add blood group, allergies, chronic conditions | profiles: updated; patient_allergies: new row                  |
| Upload lab report     | Patient Profile !' Documents                   | patient_health_folders; patient_health_documents: new row      |
| View prescriptions    | Go to prescriptions tab                        | patient_prescribed_medicines: rows visible                     |
| Set medicine reminder | Add medicine !' set reminder                   | patient_prescribed_medicines: reminder_time set                |

### 3.3 Teleconsult Flow (Video call)

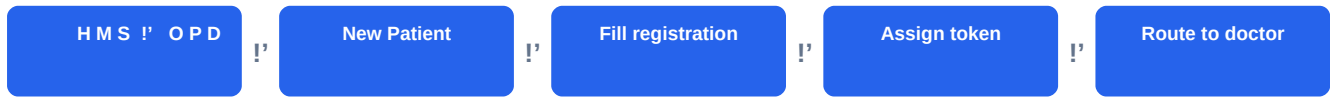


- Patient opens confirmed appointment !' clicks 'Join Video Call' !' room URL opens
- Doctor opens same appointment !' joins room !' both are now connected.
- After consultation, doctor writes prescription (Section 4.3 details this).
- Patient's dashboard shows new prescription under 'My Prescriptions'.
- DB verify: hms\_teleconsult\_sessions — status='completed', actual\_start and actual\_end populated.

## Section 4 — Doctor & HMS Flow (End-to-End)

The doctor/HMS flow is the core of clinical operations. Test the full journey from patient check-in to discharge.

### 4.1 OPD Patient Registration

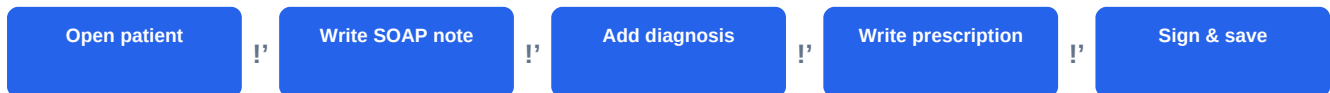


| Step | Action                                             | Expected result                          | DB table                                   |
|------|----------------------------------------------------|------------------------------------------|--------------------------------------------|
| 1    | Login as Admin/Doctor                              | OPD queue and new patient button visible | —                                          |
| 2    | Click 'New Patient' & fill: gender, phone, address | Form accepts input                       | —                                          |
| 3    | Submit registration                                | Patient ID auto-generated (e.g. AYZ-001) | hms_op_patients: 1 new row, patient_id set |
| 4    | Assign to doctor, issue token                      | Token number assigned                    | hms_op_visits: 1 new row, token_number set |
| 5    | Patient appears in today's queue                   | Queue shows patient with waiting status  | opd_queue_entries: status='waiting'        |

### 4.2 Triage (Vitals Capture)

- Open patient in OPD queue & click 'Capture Vitals' or 'Triage'.
- Fill: BP (120/80), Pulse (72), Temperature (98.6°F), Weight (70kg), SpO2 (98%).
- Add chief complaint: 'Lower back pain, 3 months duration'.
- Save triage & DB: hms\_triage\_records — 1 new row with all vitals populated.
- Patient status updates to 'Vitals Captured' in queue.

### 4.3 Clinical Consultation & Prescription



| Step              | Action                                                                 | DB table + field                                         |
|-------------------|------------------------------------------------------------------------|----------------------------------------------------------|
| Open patient      | Click patient in queue & 'Start Consultation'                          | hms_op_visits: status='in_consultation'                  |
| SOAP notes        | Fill Subjective, Objective, Assessment, Plan fields                    | hms_clinical_notes: 1 new row, note_type='soap'          |
| Ayurveda fields   | Add dosha assessment, prakriti notes, agni/ama status                  | hms_clinical_notes: dosha_assessment, agni_status fields |
| Diagnosis         | Type diagnosis + ICD/NAMASTE code                                      | hms_prescriptions: diagnosis field                       |
| Add medicine      | Medicine name, dosage (1-0-1), duration (7 days), anupana (warm water) | hms_prescription_items: 1+ new rows                      |
| Follow-up date    | Set follow-up in 7 days                                                | hms_prescriptions: follow_up_date                        |
| Sign prescription | Click 'Sign & Issue'                                                   | hms_prescriptions: is_signed=true, signed_at=now()       |
| Complete visit    | Click 'Complete Consultation'                                          | hms_op_visits: status='completed', check_out_time set    |

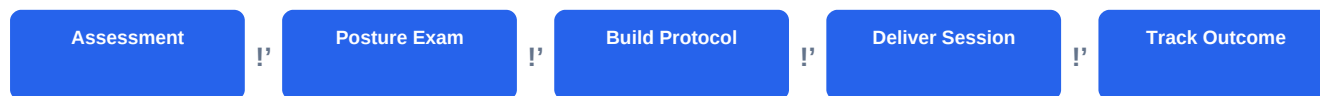
### 4.4 Billing

| Step                  | Action                                              | DB table                                           |
|-----------------------|-----------------------------------------------------|----------------------------------------------------|
| <b>Create bill</b>    | H M S !' B i l l i n g !' N e w B i l l !' a        | hms_bills: 1 new row, bill_number auto-generated   |
| <b>Add line items</b> | Consultation fee, lab, medicines                    | hms_bill_items: rows per item                      |
| <b>Set payment</b>    | Mark as Cash/UPI/Card, enter amount                 | hms_bills: payment_status='paid', payment_mode set |
| <b>Issue receipt</b>  | Click 'Issue Receipt'                               | hms_payment_receipts: 1 new row                    |
| <b>Shift closing</b>  | E n d o f d a y !' S h i f t C l o s i n g<br>tally | hms_shift_closings: 1 new row, difference=0        |

## Section 5 — Spine AYUSH (End-to-End)

The Spine AYUSH module is the core clinical differentiator. Test the full 5-step Treat!' Track.

### 5.1 The 5-Step Spine Clinical Flow



### 5.2 Step-by-step test

| Step                 | Where                              | Action                                                                               | DB table to verify                               |
|----------------------|------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------|
| 1 — Assess           | HMS !' Spine !' Assessment         | Fill patient details, chief complaint, duration, VAS pain score (baseline e.g. 7/10) | spine_wellness_assessments or spine_leads        |
| 2 — Posture Exam     | HMS !' Spine !' Posture Assessment | Fill posterior/anterior/lateral findings, identify syndrome (e.g. Lower Cross)       | spine_ayush_posture_assessments: 1 new row       |
| 3 — Protocol         | HMS !' Spine !' Treatment Protocol | Assign phased protocol (e.g. post_op), set sessions, goals                           | spine_therapy_prescriptions: 1 new row           |
| 4a — Safety Check    | HMS !' Spine !' 1 Session          | Pre-treatment safety checklist must be 100% complete before proceeding               | spine_checklists: all_clear=true                 |
| 4b — Deliver Session | Level 1 Session recorder           | Record therapy, duration, body area, pain before/after, immediate response           | spine_therapy_sessions: 1 new row                |
| 4c — Checkpoints     | Session checkpoints panel          | Mark all checkpoints: preparation, execution, aftercare, safety                      | spine_therapy_sessions: checkpoints_done updated |
| 5 — Track Outcome    | HMS !' Spine !' Outcome Tracker    | Enter post-session VAS (e.g. 4/10), ROM, functional score                            | spine_recovery_scores: 1 new row                 |
| Verify improvement   | Outcome chart                      | Chart shows VAS improvement from across sessions                                     | spine_recovery_scores: vas_pain trend            |

### 5.3 Tele-Rehab Follow-Up (Spinal Injury)

- HMS !' Spine !' Rehab Registry !' Enroll patient with injury type 'post\_op', injury 'L4-L5', phase 'sub\_acute'.
- DB verify: spine\_injury\_patients — 1 new row, assigned\_protocol set.
- Go to Tele-Rehab Follow-Up !' select patient !' record week 1 check-in.
- Tick red flag: 'New neurological deficit appearing' !' needs\_referral should aut
- DB verify: spine\_rehab\_followups — needs\_referral=true, red\_flags array populated.
- Verify patient's current\_mobility is updated in spine\_injury\_patients.

### 5.4 CSR Fund Test

- HMS !' Spine !' CSR Fund & Sponsorship !' confirm demo donor (15,00,000 grant
- Allocate 120,000 from donor to an enrolled injury patient.
- DB verify: spine\_csr\_allocations — 1 new row; spine\_csr\_donors — allocated\_amount increased.
- Check patient row: spine\_injury\_patients — is\_csr\_sponsored=true.
- Public page test: open ayuzee.com/spine/impact (no login) — aggregate totals show, NO patient names visible.

## Section 6 — Shop & Checkout (End-to-End)

Browse shop



Add to cart



Checkout



Razorpay TEST



Order confirmed



DB verified

### 6.1 Full checkout test (Razorpay TEST mode)

#### Razorpay test card details

Card: 4111 1111 1111 1111 | Expiry: Any future date (e.g. 12/30)

CVV: 123 | Name: Any name | OTP: 1234

Use these ONLY in test mode. Never use real card details in testing.

| Step | Action                                                      | Expected result                         | DB verify                                              |
|------|-------------------------------------------------------------|-----------------------------------------|--------------------------------------------------------|
| 1    | Login as patient   go to /shop                              | Product grid with categories visible    | —                                                      |
| 2    | Click a product   'Add to Cart'                             | Cart icon shows count = 1               | —                                                      |
| 3    | Go to /cart                                                 | Cart shows product, quantity, total     | —                                                      |
| 4    | Click 'Checkout'                                            | Address form appears                    | —                                                      |
| 5    | Fill delivery address (name, phone, address, city, pincode) | Form accepts input                      | —                                                      |
| 6    | Click 'Pay Now'                                             | Razorpay checkout modal opens           | orders: 1 new row, status='pending'                    |
| 7    | In Razorpay: select Card, fill test card details            | Payment processing spinner              | —                                                      |
| 8    | Enter OTP: 1234                                             | Payment accepted                        | orders: payment_status='paid', razorpay_payment_id set |
| 9    | Redirect to order success page                              | 'Thank you' or order confirmation shown | orders: order_status='confirmed'                       |
| 10   | Check dashboard   My Orders                                 | New order appears with order ID         | orders: row with correct user_id                       |

### 6.2 Prescription upload (shop)

- Go to /shop | find a prescription-required product | add to cart.
- At checkout: 'Upload Prescription' prompt appears.
- Upload a PDF/image | click Submit.
- DB verify: Supabase Storage 'prescriptions' bucket — file uploaded.
- orders: prescription item present in items JSON.

## Section 7 — Admin & Super Admin Testing

### 7.1 Admin access verification (Dr Sindhu Roja)

| Test                      | Steps                                                  | Pass condition                                                             |
|---------------------------|--------------------------------------------------------|----------------------------------------------------------------------------|
| All HMS modules visible   | Login as admin !' go to /h sidebar                     | All modules (OPD, Clinical, Lab, Stock, Accounts, MIS, Spine etc.) visible |
| Owner pages blocked       | Try to navigate to Spine Super Admin page              | Redirected / access denied — admin cannot see owner-only pages             |
| Data write access         | Create a new patient in OPD                            | Patient saved to hms_op_patients; admin sees it                            |
| Admin can read all tables | Open MIS dashboard                                     | Revenue, patient stats, all branches data visible                          |
| Cannot change roles       | No 'Role Management' or 'Grant Super Admin' UI visible | Super Admin cannot be granted by admin role                                |

### 7.2 Super Admin access verification (Owner)

- Login as curesure4u@gmail.com (super\_admin) !' HMS !' Spine section.
- Spine Super Admin page: accessible and shows all configuration options.
- Chief Physician page: accessible, shows full clinical overview.
- Platform Admin: can see all users, all roles, audit logs.
- Can grant/revoke HMS access to doctors via Access Requests.

### 7.3 Role boundary tests

| Scenario                        | User                   | Expected                                 |
|---------------------------------|------------------------|------------------------------------------|
| Patient tries to access /hms    | Patient                | Redirected away or access denied         |
| Doctor tries Spine Super Admin  | Doctor (branch_doctor) | Access denied page                       |
| Admin tries Spine Super Admin   | Admin                  | Access denied (owner-only since our fix) |
| Anon tries spine_csr_donors API | Not logged in          | Returns [] — no data exposed             |
| Anon calls public impact RPC    | Not logged in          | Returns aggregate totals only — no PII   |

## Section 8 — Notifications & Cross-Table Writes

This is the most important section. Every user action must trigger the correct DB writes AND notify the right people. This is what makes the app a real-time system.

### 8.1 Notification flow map

| Trigger event               | Who is notified  | How                                     | Tables written                                                                   |
|-----------------------------|------------------|-----------------------------------------|----------------------------------------------------------------------------------|
| Patient books appointment   | Doctor           | In-app notification + (future) WhatsApp | appointments; unified_notifications (doctor)                                     |
| Doctor issues prescription  | Patient          | In-app notification                     | hms_prescriptions; patient_prescribed_medicines; unified_notifications (patient) |
| Lab result uploaded         | Doctor + Patient | In-app alert                            | hms_lab_results; hms_lab_critical_alerts (if critical); bridge_lab_report_push   |
| Spine session completed     | Patient          | Follow-up reminder scheduled            | spine_therapy_sessions; follow_up_reminders                                      |
| Red flag triggered in rehab | Referring doctor | Urgent notification                     | spine_rehab_followups: needs_referral=true; unified_notifications                |
| New CSR allocation          | Admin            | Internal log                            | spine_csr_allocations; spine_csr_donors (updated amounts)                        |
| Package nearing expiry      | Patient          | Reminder                                | spine_patient_subscriptions; unified_notifications                               |
| New referral application    | Owner            | In-app                                  | spine_referral_applications: status='new'                                        |

### 8.2 Testing notification delivery

- Step 1: Open the Supabase Dashboard ! Table Editor ! unified\_notifications. L
- Step 2: As a patient, book an appointment with a doctor.
- Step 3: Refresh unified\_notifications — a new row must appear with user\_id = doctor's ID, type='appointment\_upcoming'.
- Step 4: Login as the doctor in a separate browser tab — the notification bell should show the new appointment.
- Step 5: Repeat for each trigger in the table above. DB row = notification was sent.

#### Why this matters

A UI that shows 'Booking confirmed' but doesn't write to the appointments table or trigger a notification is a silent failure. Always check the DB after every action. A confirmed appointment with no DB row = the booking was lost.

### 8.3 WhatsApp notifications (manual test — current state)

- Currently: follow-up buttons generate a click-to-WhatsApp link (opens WhatsApp app with pre-filled message).
- Test: HMS ! Spine ! Follow-Up Rules ! click 'Send Follow-up' for a patient ! W correct patient name and message pre-filled.
- Future: When WhatsApp Business API is connected, this will fire automatically without manual click.

## Section 9 — Security, Performance & Accessibility

### 9.1 Security tests

| Test                      | How to test                                                                          | Pass condition                                       |
|---------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------|
| RLS blocks anon           | Open browser DevTools !' Network !' GET /rest/v1/spine_injury_patients with anon key | Returns [] — no patient data                         |
| Protected routes redirect | Without logging in, go to /hms                                                       | Redirects to /auth login page                        |
| No secrets in page source | View page source (Ctrl+U) on ayuzee.com                                              | No API keys, passwords, or service role keys visible |
| XSS prevention            | In any search box, type: <script>alert('xss')</script>                               | No alert popup. Input treated as text.               |
| HTTPS only                | Open http://ayuzee.com                                                               | Auto-redirects to https://ayuzee.com                 |
| Security headers          | Open DevTools !' Network !' request !' Response Headers                              | x-frame-options, x-content-type-options present      |

### 9.2 Performance benchmarks

| Page                                   | Target load time | How to measure                                 |
|----------------------------------------|------------------|------------------------------------------------|
| Home (/)                               | < 3 seconds      | Chrome DevTools !' Network !' DOMContentLoaded |
| HMS Dashboard (/hms)                   | < 4 seconds      | Same — after login                             |
| Spine AYUSH (/hms/spine-ayush-modules) | < 3 seconds      | Same                                           |
| Shop (/shop)                           | < 3 seconds      | Same                                           |
| Doctors (/doctors)                     | < 3 seconds      | Same                                           |

To run a full Lighthouse audit: open Chrome DevTools !' Lighthouse tab !' Run audit. Accessibility >90, Best Practices >90, SEO >90.

### 9.3 Mobile responsiveness

- Open Chrome DevTools !' Toggle device toolbar (Ctrl+Shift+M) !' select iPhone (390×844).
- Test: Home page, /auth login, /doctors, /shop, /patient/dashboard.
- All buttons must be tappable (min 44×44px), text must be readable, no horizontal scroll.
- HMS modules: check sidebar collapses to hamburger menu on mobile.

### 9.4 Accessibility quick check

- Install axe DevTools Chrome extension (free).
- Open each main page !' click axe icon !' Run Analysis.
- Zero 'Critical' or 'Serious' violations is the target.
- Common issues to look for: missing form labels, insufficient color contrast, images without alt text.

## Section 10 — Database Verification Checklist

After each manual test, run these SQL queries in the Supabase Dashboard !' SQL correctly.

### 10.1 Quick verification queries

```
-- After patient signup:
SELECT id, email, created_at FROM auth.users ORDER BY created_at DESC LIMIT 3;

-- After OPD patient registration:
SELECT patient_id, first_name, mobile, created_at FROM hms_op_patients
ORDER BY created_at DESC LIMIT 5;

-- After appointment booking:
SELECT id, patient_name, appointment_date, status, payment_status
FROM hms_appointment_bookings ORDER BY created_at DESC LIMIT 5;

-- After prescription issued:
SELECT id, patient_name, doctor_name, diagnosis, is_signed, signed_at
FROM hms_prescriptions ORDER BY created_at DESC LIMIT 5;

-- After spine therapy session:
SELECT id, patient_id, session_date, therapy_name, pain_before, pain_after, status
FROM spine_therapy_sessions ORDER BY created_at DESC LIMIT 5;

-- After notification triggered:
SELECT id, user_id, title, type, is_read, created_at
FROM unified_notifications ORDER BY created_at DESC LIMIT 10;

-- After billing:
SELECT bill_number, patient_name, total_amount, payment_status, payment_mode
FROM hms_bills ORDER BY created_at DESC LIMIT 5;

-- Verify anon cannot read patient data (should return 0):
SELECT COUNT(*) FROM hms_op_patients; -- run as anon JWT to verify RLS
```

### 10.2 What a passing test looks like

| Test scenario       | DB rows that must exist after test                                                     |
|---------------------|----------------------------------------------------------------------------------------|
| Patient signed up   | auth.users: new row; profiles: new row; user_roles: role='patient'                     |
| Appointment booked  | appointments OR hms_appointment_bookings: 1 new row with correct patient + doctor IDs  |
| OPD visit completed | hms_op_patients: 1 new row; hms_op_visits: 1 new row; hms_triage_records: vitals row   |
| Prescription issued | hms_prescriptions: is_signed=true; hms_prescription_items: 1+ rows                     |
| Spine session done  | spine_therapy_sessions: 1 new row; spine_checklists: all_clear=true (if Level 1)       |
| Payment completed   | orders OR hms_bills: payment_status='paid'; razorpay_payment_id/reference set          |
| Notification sent   | unified_notifications: new row for recipient user_id                                   |
| Lab report pushed   | bridge_lab_report_push: push_status='pushed_to_app'; patient_health_documents: new row |

## Section 11 — Running Automated Tests (Playwright)

Your project already has Playwright installed and 13 test specs ready. Here is how to run them.

### 11.1 Available test specs

| Spec file                       | What it tests                     | Needs credentials?  |
|---------------------------------|-----------------------------------|---------------------|
| auth.spec.ts                    | Signup, login, wrong password     | Yes (E2E_PATIENT_*) |
| booking.spec.ts                 | Book appointment with doctor      | Yes                 |
| checkout.spec.ts                | Shop cart + Razorpay test payment | Yes                 |
| doctor-prescription.spec.ts     | Doctor writes prescription        | Yes (E2E_DOCTOR_*)  |
| security.spec.ts                | Headers, XSS, no secrets in HTML  | No                  |
| performance.spec.ts             | Load times, LCP, bundle size      | No                  |
| accessibility.spec.ts           | WCAG 2.1 AA compliance (axe)      | No                  |
| broken-links.spec.ts            | No 404 links on key pages         | No                  |
| password-reset.spec.ts          | Forgot password flow              | Yes                 |
| mobile-check.spec.ts            | Mobile viewport responsive        | No                  |
| seo.spec.ts                     | Meta tags, Open Graph, sitemap    | No                  |
| suggestion-autocomplete.spec.ts | Search/autocomplete               | Partial             |
| mis-drilldown-params.spec.ts    | MIS report drill-down             | Yes                 |

### 11.2 Setup & run commands

```
# 1. Create .env.e2e with your real test credentials (see Section 1.2)

# 2. Install browsers (first time only)
npx playwright install chromium

# 3. Run ALL tests against live site
npx playwright test --project=chromium

# 4. Run only non-credential tests (safe to run anytime)
npx playwright test security performance accessibility seo mobile-check broken-links

# 5. Run a single spec
npx playwright test auth.spec.ts

# 6. Open the HTML report after running
npx playwright show-report
```

### 11.3 Reading the test report

- Green PASSED = the flow worked and the UI showed the expected result.
- Red FAILED = either the UI changed, the credentials were wrong, or there is a real bug.
- On failure: Playwright saves a screenshot + trace. Open with: `npx playwright show-report`
- Trace viewer shows every click, every network request, every DOM state — use it to pinpoint exactly where the failure happened.

#### Recommended test cadence

Daily: Run security + performance + accessibility (no credentials needed, fast).

Before every deploy: Run auth + booking + checkout + doctor-prescription.

Weekly: Run all 13 specs with fresh credentials.

## Section 12 — Bug Report Template

When you find an issue during testing, use this template to report it. A good bug report gets it fixed fast.

| Field              | What to fill                                                                               |
|--------------------|--------------------------------------------------------------------------------------------|
| Title              | Short, specific: 'Appointment booking — no row in appointments table after booking'        |
| Section            | Which section from this guide (e.g. Section 3.1 — Book Appointment)                        |
| Severity           | Critical (data lost/security) / High (feature broken) / Medium (wrong UI) / Low (cosmetic) |
| Steps to reproduce | Numbered step-by-step: 1. Login as patient 2. Go to /doctors 3. Click Book...              |
| Expected result    | What SHOULD happen: 'A new row in appointments table with status=pending'                  |
| Actual result      | What DID happen: 'No row in appointments table; UI showed confirmation toast'              |
| DB state           | Paste the SQL query output from Section 10 showing the missing/wrong data                  |
| Screenshot         | Attach the screenshot (Playwright saves these automatically on failure)                    |
| Browser + OS       | e.g. Chrome 126 on macOS 14.5                                                              |
| Reproducible?      | Yes always / Sometimes / Only once                                                         |

### Severity guide

| Severity      | Definition                               | Example                                                   | Fix timeline          |
|---------------|------------------------------------------|-----------------------------------------------------------|-----------------------|
| Ø=Ý4 Critical | Data lost, security breach, auth broken  | Patient data readable by anon; login fails for all users  | Fix immediately       |
| Ø=ßà High     | Core feature broken, wrong DB write      | Appointment booked but no DB row; prescription not saving | Fix before going live |
| Ø=ßá Medium   | Feature partially works, wrong behaviour | Wrong patient name in notification; filter not working    | Fix in next sprint    |
| Ø=ßâ Low      | Cosmetic, minor UX                       | Button colour wrong; text truncated; icon missing         | Fix when convenient   |

#### The most important rule of testing

Every test is only complete when you have verified the database. The UI can show success while the backend silently failed. Open Supabase!' SQL Editor!' run the verification query from Section 10 after EVERY test. No DB row = the test failed, regardless of what the screen shows.